The National School of Artificial Intelligence (ENSIA)

Data Mining – Extra Exercises

2025-2026

Exercise 1

Consider the following dataset that contains the age and gender information for 9 users who visited a given website.

UserID	1	2	3	4	5	6	7	8	9
Age	17	24	25	28	32	38	39	49	68
Gender	Female	Male	Male	Male	Female	Female	Female	Male	Male

1. Suppose you apply an equal interval width approach to discretize the *Age* attribute into 3 bins. Show the *userIDs* assigned to each of the 3 bins.

2. Repeat the previous question using the equal frequency approach.

3. Repeat question (1) using a supervised discretization approach (with *Gender* as a class attribute). Specifically, choose the bins so their members are as “pure” as possible.

Exercise 2

Numerous measures have been proposed to determine the semantic similarity between two words using a domain ontology such as WordNet. For example, words such as *dog* and *cat* have higher semantic similarity than *dog* and *money,* since the former refers to carnivores.

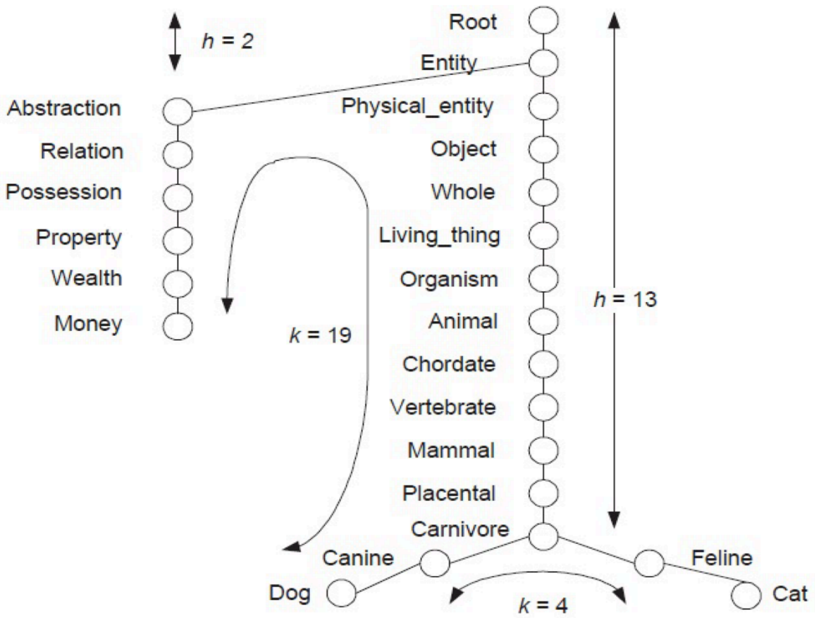

Figure Sample of the hypernym hierarchy in WordNet.

Note: In this simplified example, we assume each word has exactly one sense

This figure shows an example for computing the Wu-Palmer similarity between a *dog* and a *cat* based on their path length in the WordNet hypernym hierarchy. The depth ***h*** refers to the length of the shortest path from the root to their lowest common hypernym (e.g., *carnivore* for the word pair *dog* and *cat*), whereas ***k*** is the minimum path length between the two words

The Wu-Palmer similarity measure is defined as follows: $W = 2h / (k + 2h)$. For example, for *dog* and *cat*, $W = 26/(4 + 26) = 0.867$, whereas for *dog* and *money*, $W = 4/(19 + 4) = 0.174$.

1. What is the maximum and minimum possible value for Wu-Palmer similarity?

2. Let $1 - W$ be the Wu-Palmer distance measure. Briefly answer the following questions:

- **A.** Does $1 - W$ satisfy the positivity property, and why?
- **B.** Does $1 - W$ satisfy the symmetry property, and why?
- **C.** Does $1 - W$ satisfy the triangle inequality property, and why?

Exercise 3 (Self-work)

For each data set given below, think of examples of classification, clustering, and association rule mining.

Dataset 1: Ambulatory Medical Care Data

This dataset provides comprehensive information on ambulatory medical care, capturing detailed demographic and medical visit data for each patient. It includes variables such as gender, age, visit duration, physician diagnoses, reported symptoms, prescribed medications, and more. The dataset serves as a rich resource for analyzing patient demographics, healthcare utilization patterns, and treatment outcomes in outpatient settings.

Dataset 2: Stock Market Data

This dataset encompasses detailed stock market information, featuring historical prices and trading volumes for a variety of stocks across multiple trading days. The data includes metrics such as opening, closing, high, and low prices, along with daily trade volumes, offering insights for analyzing stock trends, market behavior, and performance over time.

Dataset 3: Premier League Football Data

This dataset contains comprehensive statistics on Major League Soccer (MLS), offering detailed information about players, teams, and past matches. It includes player profiles, team performance metrics, and in-depth game statistics such as goals, assists, possession, shots on target, fouls, and more. This data is ideal for in-depth analysis of player and team performance, game strategies, and league trends over time.